**-- Create Database**

```
CREATE DATABASE CollegeDB;
USE CollegeDB;
```

**Table 1: Department**

```
CREATE TABLE Department (
    DeptID INT PRIMARY KEY,
    DeptName VARCHAR(50) NOT NULL UNIQUE,
    HOD VARCHAR(50) NOT NULL
);
```

Insert Records

```
INSERT INTO Department VALUES
(101,'Computer Science','Dr. Rajesh Kumar'),
(102,'Information Technology','Dr. Anita Sharma'),
(103,'Electronics','Dr. Vivek Singh'),
(104,'Mechanical','Dr. Pooja Verma');
```

**Table 2: Student**

```
CREATE TABLE Student (
    StudentID INT  PRIMARY KEY,
    StudentName VARCHAR(50) NOT NULL,
    Email VARCHAR(100) UNIQUE,
    Age INT CHECK (Age >= 17),
    Gender ENUM('Male','Female','Other') NOT NULL,
    City VARCHAR(40) DEFAULT 'Delhi',
    DeptID INT,
    FOREIGN KEY (DeptID)
```

```
    REFERENCES Department(DeptID)
);

INSERT INTO Student(StudentName,Email,Age,Gender,City,DeptID)
VALUES
('Aman Gupta','aman@gmail.com',19,'Male','Delhi',101),
('Riya Sharma','riya@gmail.com',20,'Female','Noida',102),
('Karan Singh','karan@gmail.com',18,'Male','Lucknow',101),
('Neha Verma','neha@gmail.com',21,'Female','Delhi',103),
('Arjun Mehta','arjun@gmail.com',22,'Male',DEFAULT,104);
```

**Table 3: Course**

```
CREATE TABLE Course (
    CourseID INT PRIMARY KEY,
    CourseName VARCHAR(60) NOT NULL UNIQUE,
    Credits INT CHECK (Credits BETWEEN 2 AND 5)
);

Insert Records
INSERT INTO Course VALUES
(201,'Database Management System',4),
(202,'Operating System',4),
(203,'Computer Networks',3),
(204,'Data Structures',5);
```

**Table 4: Enrollment**

```
CREATE TABLE Enrollment (
    EnrollmentID INT AUTO_INCREMENT PRIMARY KEY,
    StudentID INT,
    CourseID INT,
    Semester INT CHECK (Semester BETWEEN 1 AND 8),
    Grade CHAR(2) DEFAULT 'NA',
    FOREIGN KEY(StudentID)
    REFERENCES Student(StudentID),
    FOREIGN KEY(CourseID)
    REFERENCES Course(CourseID)
);
```

Insert Records

```
INSERT INTO Enrollment(StudentID,CourseID,Semester,Grade)
VALUES
(1,201,3,'A'),
(1,202,3,'B'),
(2,201,4,'A'),
(3,204,2,'B'),
(4,203,5,'A'),
(5,202,6,DEFAULT);
```